

ASYMPTOTICALLY OPTIMAL BINARY LINEAR COVERING CODES OF RADIUS TWO

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ABSTRACT. We construct binary linear codes of covering radius at most two, with lengths tending to infinity and covering densities tending to one. Thus the asymptotic linear covering density satisfies $f(2) = 1$, answering the radius-two question in the discussion of Green's Problem 40. The construction proceeds through binary quadratic maps $Q : V \rightarrow H$, where $\dim H = \dim V - 1$. Outside an exceptional subspace of increasing codimension, every polar map $B_Q(a, \cdot)$ has kernel $\{0, a\}$. A finite interpolation operator permits the exceptional codimension to be doubled. The graph of Q then covers every required sum outside a smaller subspace; adjoining two complementary subspaces fills the remaining sums at a cost of $o(|V|)$ points. Taking these points as parity-check columns produces the codes. At iteration r the density is at most $1 + 10 \cdot 2^{-2^r}$. The construction, the density estimate, and the resulting equality have been formalized in Lean 4 and checked locally against the specified Formal Conjectures target. All proof code is supplied in one Lean file.

1. INTRODUCTION

For a nonnegative integer N , let

$$H_2(N) = \{x \in \mathbb{F}_2^N : \text{wt}(x) \leq 2\}, \quad \beta_2(N) = |H_2(N)| = 1 + N + \binom{N}{2}.$$

A binary linear code $C \leq \mathbb{F}_2^N$ has covering radius at most two if $C + H_2(N) = \mathbb{F}_2^N$. Its radius-two covering density is

$$\mu_2(C) = \frac{|C| \beta_2(N)}{2^N}. \quad (1)$$

Write

$$m_2(N) = \min_{\substack{C \leq \mathbb{F}_2^N \\ C + H_2(N) = \mathbb{F}_2^N}} \mu_2(C), \quad f(2) = \liminf_{N \rightarrow \infty} m_2(N). \quad (2)$$

The minimum exists because there are finitely many subspaces and the whole space is a covering code. The sphere-covering bound gives $m_2(N) \geq 1$.

Green's Problem 40 concerns the corresponding function $f(r)$ for general fixed covering radius r , and asks whether it tends to infinity with r . Its discussion singles out the equality $f(2) = 1$ as an unresolved case [4, Problem 40]. We prove this equality by constructing a sequence of linear codes. For general background on covering codes, see [1].

Theorem 1.1. *There is a sequence of binary linear codes $C_r \leq \mathbb{F}_2^{N_r}$, $r \geq 0$, of covering radius at most two such that $N_r \rightarrow \infty$ and*

$$1 \leq \mu_2(C_r) \leq 1 + 10 \cdot 2^{-2^r}. \quad (3)$$

In particular, $f(2) = 1$.

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The construction is quantitative. Put $p_r = 2^r$, $c_r = 2p_r$, and define

$$n_0 = 2, \quad n_{r+1} = n_r + c_r + c_r 2^{c_r} (n_r - 1). \quad (4)$$

The codes can be chosen to satisfy

$$2^{n_r} \leq N_r \leq 2^{n_r} + 2^{n_r - p_r + 1}, \quad N_r - \dim C_r = 2n_r - 1. \quad (5)$$

Their covering radius is exactly two for every $r \geq 1$.

The construction uses quadratic maps whose polar maps are surjective outside a subspace of large codimension. Their graphs supply all sums outside a smaller subspace. Two complementary subspaces supply the remaining sums, and a parity-check construction gives the codes. The rapid dimension growth in (4) is compatible with the subsequence definition (2). The behavior of $f(r)$ as $r \rightarrow \infty$ remains a separate question.

2. COVERING CODES AND SUM BASES

All vector spaces and all unqualified linear maps in the proof are over $\mathbb{F}_2$. For subsets A, B of an abelian group, write $A + B = \{a + b : a \in A, b \in B\}$. Repetitions are allowed when $A = B$.

Lemma 2.1 (Sphere-covering bound). *If $C + H_2(N) = \mathbb{F}_2^N$, then $\mu_2(C) \geq 1$.*

Proof. The addition map $C \times H_2(N) \rightarrow \mathbb{F}_2^N$ is surjective. Consequently $|C| \beta_2(N) \geq 2^N$, which is the asserted inequality. $\square$

Lemma 2.2 (Parity-check construction). *Let G be a finite binary vector space, and let $S \subseteq G$ satisfy $S + S = G$. Set $N = |S|$. There is a binary linear code $C \leq \mathbb{F}_2^N$ of covering radius at most two such that*

$$\dim C = N - \dim G, \quad \mu_2(C) = \frac{\beta_2(N)}{|G|}. \quad (6)$$

Proof. Enumerate the distinct elements of S as $s_1, \dots, s_N$, and define

$$\varphi : \mathbb{F}_2^N \longrightarrow G, \quad \varphi(x) = \sum_{i=1}^N x_i s_i.$$

Every $g \in G$ is $s_i + s_j$ for some i, j , so it equals $\varphi(e_i + e_j)$. Here $e_i + e_j$ has weight at most two, including the case $i = j$. Thus φ is surjective and every syndrome has a representative of weight at most two.

Set $C = \ker \varphi$. Given $x \in \mathbb{F}_2^N$, choose $e \in H_2(N)$ with $\varphi(e) = \varphi(x)$. Then $x - e \in C$, proving the covering property. Rank-nullity gives $|C| = 2^N / |G|$, and (6) follows. $\square$

Remark 2.3. We retain the zero element when it belongs to S , so the parity-check matrix may have a zero column. This convention simplifies the formal construction and has no effect on the limiting density. All columns are indexed by the elements of a set, so there are no repeated columns.

3. QUADRATIC MAPS WITH AN EXCEPTIONAL SUBSPACE

For a map $Q : V \rightarrow H$ with $Q(0) = 0$, its polar is

$$B_Q(a, x) = Q(a + x) + Q(a) + Q(x).$$

We call Q quadratic if B_Q is bilinear. The polar is symmetric and alternating, so $B_Q(a, a) = 0$.

Definition 3.1. An admissible quadratic map with parameters (n, c) consists of $F = \mathbb{F}_{2^c}$, binary vector spaces K, H , and a quadratic map $Q : F \oplus K \rightarrow H$ such that

$$\dim(F \oplus K) = n, \quad \dim H = n - 1, \quad (7)$$

$$Q(0, z) = 0 \quad (z \in K), \quad (8)$$

$$\ker B_Q((a, z), \cdot) = \{0, (a, z)\} \quad (a \in F \setminus \{0\}, z \in K). \quad (9)$$

We identify K with $\{0\} \oplus K$ inside $V = F \oplus K$ and call it the exceptional subspace. It has codimension c . No F -linearity is imposed on Q or on K and H .

Lemma 3.2. *For an admissible map, and every $a \in V \setminus K$, the map $x \mapsto B_Q(a, x)$ is surjective onto H . Each fiber has two elements. Consequently, if*

$$\Gamma_Q = \{(x, Q(x)) : x \in V\} \subseteq V \oplus H,$$

then $\{a\} \times H \subseteq \Gamma_Q + \Gamma_Q$ for every $a \notin K$.

Proof. By (9), the kernel has dimension one. Since $\dim V = n$ and $\dim H = n - 1$, rank-nullity gives surjectivity and the fiber size. For a prescribed $h \in H$, choose x such that $B_Q(a, x) = h + Q(a)$. Then

$$(x, Q(x)) + (a + x, Q(a + x)) = (a, h),$$

as required. $\square$

4. A FINITE INTERPOLATION OPERATOR

Let $F = \mathbb{F}_{2^c}$, with $c \geq 1$, and put $q = |F| = 2^c$. Choose binary coordinates $H \cong \mathbb{F}_2^m$. Let

$$Z = \{t : F \times \{1, \dots, m\} \rightarrow F\}.$$

This is an F -vector space of dimension qm and a binary vector space of dimension cqm . Choose a binary-linear functional $\lambda : F \rightarrow \mathbb{F}_2$ such that $\lambda(1) = 1$; extending 1 to a binary basis of F supplies one.

Define $\iota : H \rightarrow Z$, $D : Z \rightarrow Z$, and $P : Z \rightarrow H$ by

$$\iota(h)(s, j) = h_j, \tag{10}$$

$$(Dt)(s, j) = s t(s, j) + \sum_{u \in F} t(u, j), \tag{11}$$

$$(Pt)_j = \lambda \left(\sum_{s \in F \setminus \{0\}} t(s, j) \right). \tag{12}$$

Binary coordinates in (10) are viewed as elements of F . The maps ι and P are binary-linear, while D is F -linear. For $m = 0$ these definitions refer to the zero space and remain valid.

Lemma 4.1 (Interpolation). *For $\rho \in F$, $h \in H$, and $t \in Z$, the equation*

$$(D + \rho \text{ id})t = \iota(h) \tag{13}$$

holds if and only if

$$t(s, j) = \begin{cases} h_j, & s = \rho, \\ 0, & s \neq \rho. \end{cases} \tag{14}$$

In particular, $D + \rho \text{ id}$ is invertible, and every solution of (13) satisfies

$$Pt = \begin{cases} 0, & \rho = 0, \\ h, & \rho \neq 0. \end{cases} \tag{15}$$

Proof. Fix j and write $T_j = \sum_{u \in F} t(u, j)$. Equation (13) says

$$(s + \rho)t(s, j) + T_j = h_j \quad (s \in F).$$

At $s = \rho$ this gives $T_j = h_j$. At every other s , it then gives $(s + \rho)t(s, j) = 0$, so $t(s, j) = 0$. The sum identity forces $t(\rho, j) = h_j$. Conversely, these values satisfy the equation directly.

Taking $h = 0$ shows that $D + \rho \text{ id}$ is injective. It is therefore invertible on the finite-dimensional space Z . If $\rho = 0$, the sum in (12) is zero. Otherwise it is h_j , and $\lambda(h_j) = h_j$ because $h_j \in \mathbb{F}_2$ and $\lambda(1) = 1$. This proves (15). $\square$

The useful feature is the prescribed projection of the resolvent:

$$P(D + \rho \text{ id})^{-1} \iota(h) = \mathbf{1}_{\rho \neq 0} h.$$

The operator formulation gives this identity using only coordinate evaluation. Appendix A gives the extension-field version of the same interpolation mechanism.

5. DOUBLING THE EXCEPTIONAL CODIMENSION

Proposition 5.1 (Doubling). *An admissible quadratic map with parameters (n, c) gives an admissible quadratic map with parameters $(n^+, 2c)$, where*

$$n^+ = n + c + c2^c(n - 1). \quad (16)$$

Proof. Let $Q : F \oplus K \rightarrow H$ be the original map, and take Z, ι, D, P from Section 4 with $m = n - 1$. Put

$$V^+ = F \oplus F \oplus K \oplus Z, \quad H^+ = F \oplus H \oplus Z.$$

Define

$$Q^+(u, v, z, t) = (uv, Q(u, z) + Q(v, z) + P(vt), ut + D(vt) + \iota(Q(v, z))). \quad (17)$$

Field scalars act coordinatewise on Z . Each coordinate of Q^+ is quadratic over $\mathbb{F}_2$: products of two linear coordinate maps have bilinear polars, and composition with a linear map preserves this property. The new exceptional subspace is

$$K^+ = \{(0, 0, z, t) : z \in K, t \in Z\}.$$

Equation (8) gives $Q^+|_{K^+} = 0$. Since $\dim Z = c2^c(n - 1)$, we have (16), $\dim H^+ = n^+ - 1$, and $\text{codim}_{V^+} K^+ = 2c$.

It remains to prove the kernel condition. Let

$$A = (a, b, z_0, t_0) \notin K^+, \quad U = (u, v, z_1, t_1),$$

and suppose $B_{Q^+}(A, U) = 0$. Set

$$X = B_Q((a, z_0), (u, z_1)), \quad Y = B_Q((b, z_0), (v, z_1)), \quad R = bt_1 + vt_0.$$

Polarizing (17) gives precisely

$$av + bu = 0, \quad (18)$$

$$X + Y + P(R) = 0, \quad (19)$$

$$at_1 + ut_0 + D(R) + \iota(Y) = 0. \quad (20)$$

We consider the three possible cases for (a, b) .

Case 1: $a \neq 0$ and $b \neq 0$. Let $\rho = a/b$. Equation (18) gives $u = \rho v$, so $at_1 + ut_0 = \rho R$. Equation (20) becomes

$$(D + \rho \text{id})R = \iota(Y).$$

Lemma 4.1 gives $P(R) = Y$, and (19) therefore gives $X = 0$. The original kernel condition implies

$$(u, z_1) = \varepsilon(a, z_0) \quad \text{for some } \varepsilon \in \mathbb{F}_2.$$

Using (18) and $a \neq 0$, we obtain $v = \varepsilon b$. Thus $Y = 0$, since its second argument is either zero or its first argument. Invertibility of $D + \rho \text{id}$ now gives $R = 0$. As $b \neq 0$, $bt_1 + \varepsilon bt_0 = 0$ implies $t_1 = \varepsilon t_0$. Hence $U = \varepsilon A$.

Case 2: $a = 0$ and $b \neq 0$. Equation (18) gives $u = 0$. Since Q vanishes on K , we have $X = 0$. Equation (20) gives $DR = \iota(Y)$. The interpolation identity at $\rho = 0$ gives $P(R) = 0$, and (19) then forces $Y = 0$. The original kernel condition gives $(v, z_1) = \varepsilon(b, z_0)$ with $\varepsilon \in \mathbb{F}_2$. Invertibility of D gives $R = 0$, and hence $t_1 = \varepsilon t_0$. Again $U = \varepsilon A$.

Case 3: $a \neq 0$ and $b = 0$. Equation (18) gives $v = 0$. We have $Y = 0$ because both of its arguments lie in K , and $R = 0$. Equation (19) gives $X = 0$, so $(u, z_1) = \varepsilon(a, z_0)$. The remaining equation is $a(t_1 + \varepsilon t_0) = 0$, whence $t_1 = \varepsilon t_0$ and $U = \varepsilon A$.

These cases exhaust $A \notin K^+$. Conversely, 0 and A lie in the kernel because the polar is alternating. Thus the kernel is exactly $\{0, A\}$.

Finally, choose a binary-linear isomorphism $\mathbb{F}_{2^{2c}} \cong F \oplus F$ and transport the first two input coordinates along it. This identifies V^+ with $\mathbb{F}_{2^{2c}} \oplus (K \oplus Z)$, as required in Definition 3.1. The isomorphism only concerns the binary vector-space structure; all the properties already proved are preserved. $\square$

6. ITERATION AND REPAIR OF THE GRAPH

An admissible seed exists with $n = c = 1$: take $F = \mathbb{F}_2$, $K = H = 0$, and $Q = 0$. The only nonzero input is 1, and its polar kernel is the whole two-element domain, which is $\{0, 1\}$. The first application of Proposition 5.1 gives $n = c = 2$ and the map $(u, v) \mapsto uv$ from $\mathbb{F}_2^2$ to $\mathbb{F}_2$.

Starting at this two-dimensional map and iterating gives admissible maps Q_r with the parameters in (4). In particular $c_r = 2^{r+1}$ and $n_r \geq c_r$. The first parameters are

r	n_r	c_r	$\dim K_r$
0	2	2	0
1	12	4	8
2	720	8	712
3	1,473,240	16	1,473,224

These numbers follow from the dimension recurrence; the proof of the kernel property is the induction in Proposition 5.1.

Proposition 6.1 (Graph repair). *Let $Q : V \rightarrow H$ be admissible with parameters (n, c) , where $c = 2p$ and $p \geq 1$. There is a set $S \subseteq G = V \oplus H$ such that*

$$S + S = G, \quad 2^n \leq |S| \leq 2^n + 2^{n-p+1}. \quad (21)$$

Moreover, $\dim G = 2n - 1$.

Proof. The graph Γ_Q has 2^n points. By Lemma 3.2, every point outside

$$J = K \oplus H \subseteq V \oplus H$$

belongs to $\Gamma_Q + \Gamma_Q$. Since $c = 2p$,

$$\dim J = (n - 2p) + (n - 1) = 2(n - p) - 1.$$

Choose complementary subspaces A, B of J with

$$\dim A = n - p, \quad \dim B = n - p - 1.$$

These dimensions are nonnegative because $n \geq 2p$ and $p \geq 1$. Every element of J is a sum of an element of A and an element of B . Consequently

$$S = \Gamma_Q \cup A \cup B$$

satisfies $S + S = G$. Its cardinality obeys

$$2^n \leq |S| \leq 2^n + 2^{n-p} + 2^{n-p-1} \leq 2^n + 2^{n-p+1},$$

and $\dim G = n + (n - 1) = 2n - 1$. □

7. THE DENSITY ESTIMATE AND THE LIMIT

Lemma 7.1. *Suppose $p \geq 1$, $n \geq 2p$, and $2^n \leq N \leq 2^n + 2^{n-p+1}$. Then*

$$\frac{1 + N + \binom{N}{2}}{2^{2n-1}} \leq 1 + \frac{10}{2^p}. \quad (22)$$

Proof. Write $v = 2^n$ and $q = 2^p$. The assumptions give

$$2 \leq q \leq v, \quad v \leq N, \quad q(N - v) \leq 2v.$$

In particular $N \leq 2v$. Hence

$$\begin{aligned} q(N^2 - v^2) &= q(N - v)(N + v) \leq 2v \cdot 3v = 6v^2, \\ q(N + 2) &\leq v(2v + 2) \leq 4v^2. \end{aligned}$$

Adding the two inequalities yields

$$q(N^2 + N + 2) \leq (q + 10)v^2.$$

Since $2(1 + N + \binom{N}{2}) = N^2 + N + 2$ and $2 \cdot 2^{2n-1} = v^2$, division by qv^2 gives (22). $\square$

Proof of Theorem 1.1. Apply Proposition 6.1 to Q_r with $p = p_r = 2^r$, and let S_r be the resulting sum basis in $G_r = V_r \oplus H_r$. Lemma 2.2 gives a code C_r of length $N_r = |S_r|$, covering radius at most two, and redundancy $2n_r - 1$. Thus (5) holds, and

$$\mu_2(C_r) = \frac{1 + N_r + \binom{N_r}{2}}{2^{2n_r-1}} \leq 1 + 10 \cdot 2^{-p_r}$$

by Lemma 7.1. Lemma 2.1 gives the lower bound in (3).

Since $n_r \geq 2p_r = 2^{r+1}$, the lengths satisfy $N_r \geq 2^{n_r} \rightarrow \infty$. Therefore

$$1 \leq f(2) \leq \liminf_{r \rightarrow \infty} m_2(N_r) \leq \lim_{r \rightarrow \infty} \mu_2(C_r) = 1.$$

The first comparison of lower limits is valid because $N_r \rightarrow \infty$; strict monotonicity is unnecessary.

Finally, for $r \geq 1$ we have $n_r \geq 12$. Put $v = 2^{n_r}$. As in the preceding lemma, $N_r \leq 2v$, while

$$|G_r| = v^2/2 > 2v + 1 \geq N_r + 1.$$

There are at most $N_r + 1$ syndromes represented by vectors of weight at most one. Thus the code cannot have covering radius at most one. Its covering radius is exactly two. $\square$

8. LEAN FORMALIZATION

The file `Green40.lean` contains all helper lemmas and the theorem `Bounty.target` in one source file. Its only imports are the `Green 40` problem module and the challenge’s `TaskSupport` module. The exact challenge target resolves to

$$\text{True} \longleftrightarrow \text{Green40.f } 2 = 1.$$

The source definition of `Green40.f` uses the lower limit of an infimum over binary linear covering subspaces, in the extended nonnegative reals [3]. The fixed-length family is finite and nonempty, so this agrees with (2).

The implementation follows the operator construction of Sections 4–7. The main correspondence is recorded below; the names are relative to `Bounty.Green40Proof`.

Mathematical step	Lean declarations
Interpolation (Lemma 4.1)	<code>interp_support</code> , <code>interp_project</code> , <code>interp_injective</code>
Doubling (Proposition 5.1)	<code>lift_kernel</code> , <code>Model.step</code>
Iteration and graph repair	<code>models</code> , <code>Model.graph_sums</code> , <code>Model.exists_sum_basis</code>
Parity-check code	<code>parity_cover</code> , <code>parity_card</code>
Density and limit	<code>density_estimate</code> , <code>f_two_le_one</code> , <code>density_two_eq_one</code>

The formal proof uses the sufficient bound $2|H_2(N)| \leq N^2 + N + 2$ and includes the sphere-covering lower-bound lemmas from the public contribution [2], credited in the source. The target uses covering radius at most two; the observation giving exact radius two appears in the proof of Theorem 1.1.

The complete proof was checked locally with Lean 4.33.1 and Mathlib [5], using the following pinned source revisions:

Dependency	Git revision
Formal Conjectures	8432eac998110a563e03df65a28c117e97c8c142
Mathlib	0df444a360eaa60ab8c11dca51a86af692955474

The local kernel and task-inspector checks verified the exact target type. Its only axiom dependencies are `propext`, `Classical.choice`, and `Quot.sound`, all permitted by the challenge. There is no `sorryAx`

dependency or use of the conjectural source theorem as a proof. The accompanying verification record and instructions reproduce these local checks.

APPENDIX A. THE EXTENSION-FIELD FORM OF INTERPOLATION

The interpolation mechanism also admits a formulation with field multiplication in place of the operator D .

Let $F = \mathbb{F}_q$, where $q = 2^c$, and choose $L = \mathbb{F}_{q^q}$ and an element $\delta \in L$ of degree q over F . The elements

$$(\delta + \rho)^{-1} \quad (\rho \in F)$$

form an F -basis of L . Indeed, a linear relation among them, multiplied by $\prod_{\rho \in F} (\delta + \rho)$, gives a polynomial of degree at most $q - 1$ vanishing at δ . It is therefore the zero polynomial. Evaluating it at any $\rho \in F$ shows that the corresponding coefficient is zero: the surviving factor is

$$\prod_{\sigma \neq \rho} (\rho + \sigma) = 1,$$

because $\prod_{\sigma \in F} (X + \sigma) = X^q + X$ has derivative 1.

There is consequently an F -linear functional $\tilde{\ell} : L \rightarrow F$ taking the value 0 on δ^{-1} and the value 1 on every $(\delta + \rho)^{-1}$ with $\rho \neq 0$. Choose $\lambda : F \rightarrow \mathbb{F}_2$ with $\lambda(1) = 1$ and set $\ell = \lambda \circ \tilde{\ell}$. For $Z = L^m$, let ι include binary coordinates and let P apply ℓ coordinatewise. Then

$$P((\delta + \rho)^{-1} \iota(h)) = \mathbf{1}_{\rho \neq 0} h.$$

Multiplication by $\delta + \rho$ is invertible since $\delta \notin F$. Thus this construction has exactly the interpolation and injectivity properties used in Proposition 5.1. Replacing $D(t)$ by δt in (17) recovers the extension-field formula

$$Q^+(u, v, z, t) = (uv, Q(u, z) + Q(v, z) + P(vt), (u + \delta v)t + \iota(Q(v, z))).$$

Its binary auxiliary dimension is again $cq(n - 1)$. The two implementations of interpolation need not produce isomorphic operators; the doubling proof uses only the properties they share.

The two-dimensional seed can likewise be expressed as the norm $x \mapsto x^3$ from $\mathbb{F}_4$ to $\mathbb{F}_2$. In binary coordinates it is $u + v + uv$, whose polar agrees with that of uv . The operator-based formalization begins with the one-dimensional seed and obtains uv by its first doubling.

APPENDIX B. A SHARPER COUNT FOR THE REPAIRED GRAPH

An appropriate choice of the two repair subspaces gives an exact cardinality formula. This refinement also removes the zero column. The formal proof uses the simpler estimate in Proposition 6.1; the refinement is unnecessary for the limiting equality.

Write $c = 2p$, and retain $J = K \oplus H$. As $Q|_K = 0$,

$$\Gamma_Q \cap J = K \oplus \{0\}.$$

Choose $A \leq J$ of dimension $n - p - 1$ containing $K \oplus \{0\}$, and choose a complement B of dimension $n - p$. This is possible because $\dim K = n - 2p \leq n - p - 1$. Now

$$\Gamma_Q \cap A = K \oplus \{0\}, \quad \Gamma_Q \cap B = A \cap B = \{0\}.$$

Inclusion-exclusion gives, for $S = \Gamma_Q \cup A \cup B$,

$$s = |S| = 2^n + 3 \cdot 2^{n-p-1} - 2^{n-2p} - 1. \quad (23)$$

As before, $S + S = G$. Use the $s - 1$ nonzero elements of S as parity-check columns. Each syndrome remains the sum of at most two columns: a sum involving zero uses at most one column. The resulting

code C' has length $s - 1$, redundancy $2n - 1$, and density

$$\begin{aligned}\mu_2(C') &= \frac{1 + (s - 1) + \binom{s-1}{2}}{2^{2n-1}} \\ &= \left(\frac{s}{2^n}\right)^2 - \frac{s}{2^{2n}} + \frac{1}{2^{2n-1}}.\end{aligned}\tag{24}$$

Equation (23) gives

$$\frac{s}{2^n} = 1 + \frac{3}{2} 2^{-p} - 2^{-2p} - 2^{-n}.$$

Along the constructed sequence, $p \rightarrow \infty$ and $n \geq 2p$, so (24) supplies another direct proof that the densities tend to one. For example, the member with $(n, c) = (12, 4)$ has $s = 5375$; it gives a code of length 5374 and dimension 5351.

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REFERENCES

- [1] G. Cohen, I. Honkala, S. Litsyn and A. Lobstein, *Covering Codes*, North-Holland Mathematical Library **54**, Elsevier, Amsterdam, 1997.
- [2] Conjectures.io contributor, *Green 40 sphere-covering lower bounds*, Lean source, contribution 61648655993994f3..., 2026. Conjectures.io contribution repository, accessed 15 September 2026.
- [3] The Formal Conjectures Authors, *The Formal Conjectures Repository*, 2025, FormalConjectures/GreensOpenProblems/40.lean, revision 8432eac99811..., 2026. <https://github.com/google-deepmind/formal-conjectures>. Task: Green 40, $f(2) = 1$, accessed 15 September 2026.
- [4] B. Green, *100 Open Problems*, December 2025 revision, Problem 40, pp. 20–21. <https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf>, accessed 15 September 2026.
- [5] The Mathlib Community, *Mathlib: the math library of Lean 4*, revision 0df444a360ea..., 2026. <https://github.com/leanprover-community/mathlib4>.

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